

Classes — Player and Enemy Objects

Day 02

Lesson 02

Nora XIE

Course 02

Phase 1 — Game Studio

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Use a class to bundle name, HP, and actions
- ★ Create objects with `Player(...)`
- ★ Call methods like `player.take_damage(3)`



Course 02 quest map: Day 02 of 18

Hook

Variables scattered everywhere get messy. A `Player` object keeps your hero in one package.

Learn



Learn

A **class** is a blueprint; an **object** is one built thing.
`Player(name, hp)` stores data; `take_damage(3)` changes `self.hp`.
`self` means “this object.”



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — Run — Investigate — Modify —> Make



Demo code

Folder: day02/

```
1 class Player:
2     def __init__(self, name, hp):
3         self.name = name
4         self.hp = hp
5         self.poison_uses = 1
6
7     def take_damage(self, amount):
8         self.hp = max(0, self.hp - amount)
9
10    def is_alive(self):
11        return self.hp > 0
12
13 hero = Player("Nora", 20)
14 hero.take_damage(5)
15 print(hero.name, "HP:", hero.hp)
```

 Try it

In terminal, `cd course02/day02`, then predict output, then run: `python demos/demo_player_class.py`

 Assignment

Work in: `day02/assignment/`

- A — **Predict** After `p = Player("A", 10)` and `p.take_damage(3)`, what is `p.hp`?
- B — **Debug** `assignment/buggy_player.py` — find the `self` mistake.
- C — **Level up** Add `Player` and `Dragon` classes to your multi-file game from Day 1.
- D — **Write** One sentence: what is the difference between a class and an object?

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Save and load game state with JSON.